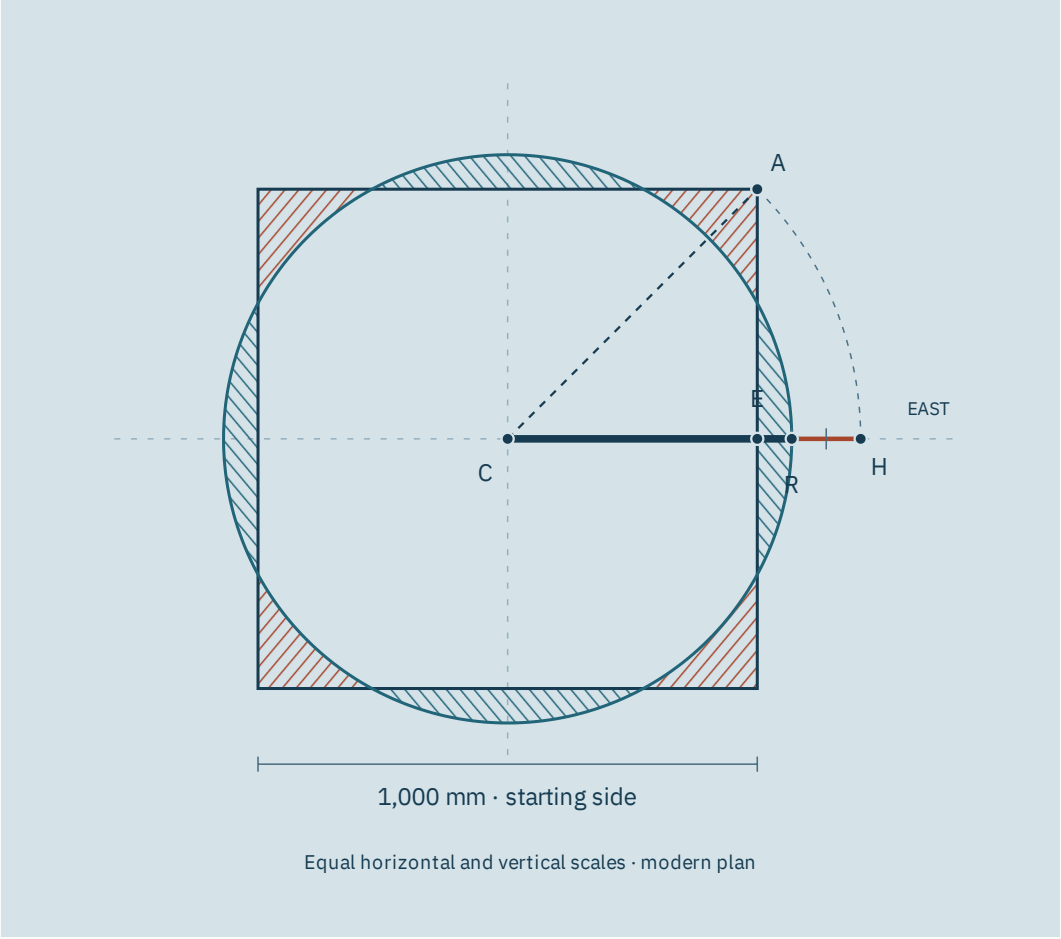
RAJJU · रज्जु · geometry by the cord

A circle too large. A return almost perfect.

Starting side: 1,000 mm, a chosen modern scale. Baudhāyana tradition, read through George Thibaut's 1875 print.

Edition 1603072da5628220cd15a11096d1c77370f46fa2223cb984c9841f6b2c8d0b9a. Open this construction online. This file needs no network to read or print.

Radius rule: source third. Selected construction step: 4 of 4. The area measurements below describe the completed circle even if an earlier step is selected.

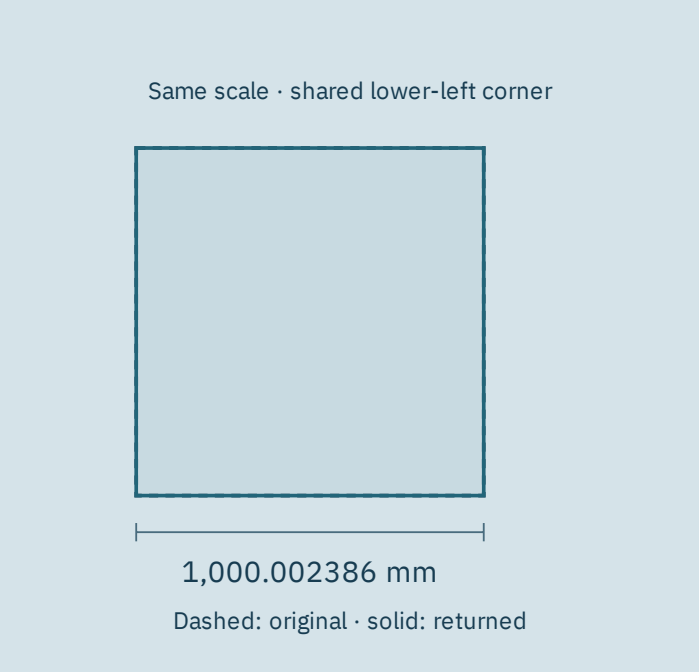


Radius = 500 + 1/3 × 207.106781187 = 569.035593729 mm.

Circle versus starting square: +1.725243539%. Corners removed: 84,051.799066 mm²; bulges added: 101,304.234459 mm².

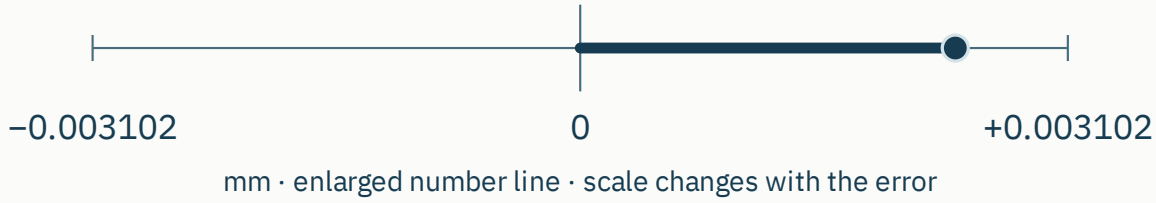
Return the circle to a square

The detailed source rule retains 9785/11136 of the circle’s diameter.



Returned square versus starting square: +0.000477179%. Returned square versus intermediate circle: -1.695514605%. A close return does not establish equal areas at each step.

The returned side differs by +0.00238589 mm. The following number line enlarges the error; the construction drawings retain equal axis scales.



Measurements and their references

Quantity	Value	Unit	Reference
selection · lesson	circle		Selected lesson
selection · side	1000	mm	Modern illustrative scale
selection · circle_rule	third		Source rule
selection · return_rule	detailed		Source rule

Quantity	Value	Unit	Reference
circle · kept_excess_fraction	0.3333333333333333	ratio	Exactly 1/3; decimal rounded
circle · half_diagonal	707.1067811865474	mm	Geometric diagonal, not 577/408
circle · excess	207.10678118654758	mm	Half diagonal minus half side
circle · radius	569.0355937288491	mm	Half side plus kept excess
circle · diameter	1138.0711874576982	mm	Twice radius
circle · original_area	1000000	mm^2	Starting square
circle · circle_area	1017252.4353936727	mm^2	Pi times radius squared
circle · lost_area	84051.79906568605	mm^2	Square outside circle
circle · gained_area	101304.23445935876	mm^2	Circle outside square
circle · area_error_percent	1.7252435393672716	%	Circle versus starting square
return · factor	9785/11136	exact ratio	Returned side divided by circle diameter
return · side	1000.0023858902279	mm	Diameter times return factor
return · area	1000004.7717861482	mm^2	Returned square
return · side_error	0.0023858902278295346	mm	Returned side minus starting side
return · area_error_percent	0.0004771786148216606	%	Returned square versus starting square
return · circle_area_error_percent	-1.6955146045779412	%	Returned square versus intermediate circle

Read the instructions

From square to circle

Baudhāyana, circle rule · Thibaut p. 26

चतुरस्रं मण्डलं चिकीर्षन्नक्ष्णयार्धं मध्यात्प्राचीमभ्यापातयेत् । यदतिशिष्यते तस्य सह तृतीयेन मण्डलं परिलिखेत् ।

caturasraṃ maṇḍalaṃ cikīrṣann akṣṇayārdhaṃ madhyāt prācīm abhyāpātayet. yad atiśiṣyate tasya saha tṛtīyena maṇḍalaṃ parilikhet.

Wishing to make a square into a circle, lay out half the diagonal from the centre toward the east. Draw the circle with a third of what projects beyond [added to the part inside].

The bracketed words make the geometrical reading explicit. They are supplied in this new English translation, not additional Sanskrit. The circle is intended as an area transformation; the modern calculation below finds a mismatch.

From circle to square

Baudhāyana, detailed return rule · Thibaut pp. 27–28

मण्डलं चतुरस्रं चिकीर्षन् विष्कम्भमष्टौ भागान्कृत्वा भागमेकोनत्रिंशद्धा विभज्याष्टाविंशतिभागानुद्धरेत् । भागस्य च षष्ठमष्टमभागोनम् ।

maṇḍalaṃ caturasraṃ cikīrṣan viṣkambham aṣṭau bhāgān kṛtvā bhāgam ekonatriṃśaddhā vibhajyāṣṭāviṃśatibhāgān uddharet. bhāgasya ca ṣaṣṭham aṣṭamabhāgonam.

Wishing to make a circle into a square, divide the diameter into eight parts, then one part into twenty-nine. Remove twenty-eight of those parts, and also a sixth of the remaining part, diminished by its eighth.

The retained side is (7/8 + 1/232 − 1/1392 + 1/11136) times the diameter. This follows Thibaut’s explicit fraction expansion. His pp. 27–28 discuss the connection to the outward transformation; a close return is not an exact equality of areas.

A shorter return rule

Baudhāyana, alternative return rule · Thibaut p. 28

अपि वा पञ्चदश भागान्कृत्वा द्वावुद्धरेत्सैषानित्या चतुरस्रकरणी ।

api vā pañcadaśa bhāgān kṛtvā dvāv uddharet. saiṣānityā caturasrakaraṇī.

Or, having divided [the diameter] into fifteen parts, remove two. This gives an inexact square-making measure.

Thibaut renders the resulting side as ‘gross’. The reading anityā is retained here, following the printed Baudhāyana passage; the neighboring authors’ formulations and interpretations differ. The numerical instruction is unambiguous: retain 13/15 of the diameter.

The numerical diagonal

Baudhāyana, diagonal rule · Thibaut pp. 12–13

प्रमाणं तृतीयेन वर्धयेत्तच्चतुर्थेनात्मचतुस्त्रिंशोनेन । सविशेषः ।

pramāṇaṃ tṛtīyena vardhayet tac caturthenātmacatustriṃśonena. saviśeṣaḥ.

Increase the measure by its third, and that [third] by its fourth, diminished by its own thirty-fourth. [The result is] saviśeṣa.

The English supplies the nested references in brackets. Thibaut explains the diagonal interpretation on p. 13. The sūtra gives the rule, not the conjectural derivation he proposes afterward. Our exact fraction calculation does not depend on that conjecture.

Sources and limits

These are modern mathematical drawings of rules transmitted under Baudhāyana’s name, not a surveyed altar or a reconstruction of an ancient building technique. Millimetres are a chosen modern scale. Sanskrit spelling, spacing and punctuation are normalized from the 1875 print; transliteration is not a pronunciation reconstruction. The source’s compressed instructions, later commentaries and Thibaut’s historical conjectures have different evidential status. No date of invention, priority claim, or route of transmission is established here.

The circle workshop uses the geometric diagonal. The numerical-diagonal lesson is separate. Exact rational expressions are distinct from rounded modern decimal comparisons.

The complete page images follow. Complete 1875 volume. The editor’s historical conjectures and judgments remain visible as source evidence, not as established findings. The nine JPEGs and all fonts were checked against the edition’s SHA-256 identities before this file was created.

Code, new drawings and English notes: MIT, Anton Soloviev, 2026. Historical source text and print images are public domain. Font licenses are retained below.

parts had to remain unchanged. A look at the outlines of the different chitis is sufficient to show that all this could not be accomplished without a certain amount of geometrical knowledge. Squares had to be found which would be equal to two or more given squares, or equal to the difference of two given squares; oblongs had to be turned into squares and squares into oblongs; triangles had to be constructed equal to given squares or oblongs, and so on. The last task and not the least was that of finding a circle, the area of which might equal as closely as possible that of a given square.

Nor were all these problems suggested only by the substitution of the more complicated forms of the agni for the primitive chaturasras'yena, although this operation doubtless called for the greatest exertion of ingenuity; the solution of some of them was required for the simplest sacrificial constructions. Whenever a figure with right angles, square or oblong, had to be drawn on the ground, care had to be taken that the sides really stood at right angles on each other; for would the áhavaníya fire have carried up the offerings of the sacrificer to the gods if its hearth had not the shape of a perfect square? There was an ancient precept that the vedi at the sautrámaṇí sacrifice was to be the third part of the vedi at the soma sacrifices, and the vedi at the piṭriyajna its ninth part; consequently a method had to be found out by which it was possible to get the exact third and ninth part of a given figure. And when, according to the opinion of some theologians, the gárhapatya had to be constructed in a square shape, according to the opinion of others as a circle, the difference of the opinions referred only to the shape, not to the size, and consequently there arose the want of a rule for turning a square into a circle.

The results of the endeavours of the priests to accomplish tasks of this nature are contained in the paribhásha sútras of the S'ulvasútras. The most important among these is, to use our terms, that referring to the hypotenuse of the rectangular triangle. The geometrical proposition, the discovery of which the Greeks ascribed to Pythagoras, was known to the old ácháryas, in its essence at least. They express it, it is true, in words very different from those familiar to us; but we must remember that they were interested in geometrical truths only as far as they were of practical use, and that they accordingly gave to them the most practical expression. What they wanted was, in the first place, a rule enabling them to draw a square of double the size of another square, and in the second place a rule teaching how to draw a square equal to any two given squares, and according to that want they worded their knowledge. The result is, that we have two propositions instead of one, and that these propositions speak of squares and oblongs instead of the rectangular triangle.

The editor's account of area transformations and their ritual setting. His claims about ancient motivations remain interpretation.

George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.

These propositions are as follows :

Baudháyana :

समचतुरस्रस्याक्ष्णयारज्जुर्द्विस्तावतीं भूमिं करोति ।

The cord which is stretched across—in the diagonal of—a square produces an area of double the size.

That is : the square of the diagonal of a square is twice as large as that square.

Ápastamba :

चतुरस्रस्याक्ष्णयारज्जुर्द्विस्तावतीं भूमिं करोति ।

Kátyáyana :

समचतुरस्रस्याक्ष्णयारज्जुर्द्विकरणी ।

The cord in the diagonal of a square is the cord (the line) producing the double (area).

"Samachaturasra" is the term employed throughout in the S'ulva-sútras to denote a square, the "sama" referring to the equal length of the four sides and the chaturasra implying that the four angles are right angles. The more accurate terminology of later Indian geometry distinguishes two classes of samachaturas'ras, or samachaturbhujas, *viz.* the samakarṇa samachaturbhuja and the vishamakarṇa samachaturbhuja ; the S'ulvasútras, having to do only with the former one, make no such distinction. Akshṇayárajju is the ancient term, representing the later "karṇarajju" or simply "karna." "Area" is here denoted by "bhúmi," while in later times "kshetra" expressed this idea, and "bhúmi" became one of the words for the base of a triangle or any other plane figure.

The side of a square is said to produce that square (karoti), a way of speaking apparently founded on the observation that the square is found by multiplying the number which expresses the measure of the side by itself ; if the side was five feet long, the square was found to consist of 5×5 little squares, &c. The expression was not applicable to other plane figures, to an oblong for instance ; for there the area is the product of two sides of different length, neither of which can be said to produce the figure by itself.

The side of a square, or originally the cord forming the side of a square, is therefore called the "karaṇí" of the square. That "rajju" is to be supplied to "karaṇí", is explicitly stated by Kátyáyana :

करणी तत्करणी तिर्यङ्मानी पार्श्वमान्यक्ष्णयेति रज्जवः ।

By the expressions : karaṇí, karaṇí of that (of any square) &c., we mean cords.

The side of a square being called its karaṇí, the side of a square of double the size was the "dvikaraṇí", the line producing the double (I shall for convenience sake often employ the terms "side" or "line"

The square's diagonal as a cord that produces double the area; Sanskrit and Thibaut's translation. George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.

एतावन्ति विज्ञेयानि वेदिविहरणानि भवन्ति ।

So many "cognizable" measurements of the vedi exist.

That means: these are the measurements of the vedi effected by oblongs, of which the sides and the diagonal can be known, *i. e.*, can be expressed in integral numbers.

In this manner A'pastamba turns the Pythagorean triangles known to him to practical use (the fourth of those which Baudháyana enumerates is not mentioned, very likely because it was not quite convenient for the measurement of the vedi), but after all Baudháyana's way of mentioning these triangles as proving his proposition about the diagonal of an oblong is more judicious. It was no practical want which could have given the impulse to such a research—for right angles could be drawn as soon as one of the "vijneya" oblongs (for instance that of 3, 4, 5) was known—but the want of some proof which might establish a firm conviction of the truth of the proposition.

The way in which the Sútrakáras found the cases enumerated above, must of course be imagined as a very primitive one. Nothing in the sútras would justify the assumption that they were expert in long calculations. Most likely they discovered that the square on the diagonal of an oblong, the sides of which were equal to three and four, could be divided into twenty-five small squares, sixteen of which composed the square on the longer side of the oblong, and nine of which formed the area of the square on the shorter side. Or, if we suppose a more convenient mode of trying, they might have found that twenty-five pebbles or seeds, which could be arranged in one square, could likewise be arranged in two squares of sixteen and of nine. Going on in that way they would form larger squares, always trying if the pebbles forming one of these squares could not as well be arranged in two smaller squares. So they would form a square of 36, of 49, of 64, &c. Arriving at the square formed by $13 \times 13 = 169$ pebbles, they would find that 169 pebbles could be formed in two squares, one of 144 the other of 25. Further on 625 pebbles could again be arranged in two squares of 576 and 49, and so on. The whole thing required only time and patience, and after all the number of cases which they found is only a small one.

Having found that, in certain cases at least, it was possible to express the sides and the diagonal of an oblong in numbers, the Sútrakáras naturally asked themselves if it would not be possible to do the same thing for a square. As the side and the diagonal of a square are in reality incommensurable quantities we can of course only expect an approximative value; but their approximation is a remarkably close one.

Baudháyana:

प्रमाणं तृतीयेन वर्धयेत्तच्च चतुर्थेनात्मचतुस्त्रिꣳशोनेन । सविशेषः ।

The numerical diagonal rule appears at the bottom of the page. The historical speculation above it is Thibaut's. George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.

Increase the measure by its third part and this third by its own fourth less the thirty-fourth part of that fourth; (the name of this increased measure) is savis'esha.

Á pastamba gives the rule in the same words.

Kátyáyana:

करणीं तृतीयेन तृतीयं चतुर्थेनात्मचतुस्त्रिंशेनेन सविशेष इति विशेषः।

The sútras themselves are of an enigmatical shortness, and do not state at all what they mean by this increasing of the measure; but the commentaries leave no doubt about the real meaning; the measure is the karaṇí, the side of a square and the increased measure the diagonal, the dvikaraṇí. If we take 1 for the measure, and increase it as directed, we get the following expression: $1 + \frac{1}{3} + \frac{1}{3 \times 4} - \frac{1}{3 \times 4 \times 34}$ and this turned into a decimal fraction gives: 1·4142156 ..... Now the side of a square being put equal to 1, the diagonal is equal to $\sqrt{2} = 1{\cdot}414213$ .. Comparing this with the value of the savis'esha we cannot fail to be struck by the accuracy of the latter.

The question arises: how did Baudháyana or Á pastamba or whoever may have the merit of the first investigation, find this value? Certainly they were not able to extract the square root of 2 to six places of decimals; if they had been able to do so, they would have arrived at a still greater degree of accuracy. I suppose that they arrived at their result by the following method which accounts for the exact degree of accuracy they reached.

Endeavouring to discover a square the side and diagonal of which might be expressed in integral numbers they began by assuming two as the measure of a square's side. Squaring two and doubling the result they got the square of the diagonal, in this case = eight. Then they tried to arrange eight, let us say again, eight pebbles, in a square; as we should say, they tried to extract the square root of eight. Being unsuccessful in this attempt, they tried the next number, taking three for the side of a square; but eighteen yielded a square root no more than eight had done. They proceeded in consequence to four, five, &c. Undoubtedly they arrived soon at the conclusion that they would never find exactly what they wanted, and had to be contented with an approximation. The object was now to single out a case in which the number expressing the square of the diagonal approached as closely as possible to a real square number. I subjoin a list, in which the numbers in the first column express the side of the squares which they subsequently tried, those in the second column the square of the diagonal, those in the third the nearest square number.

English explanation and fraction expansion for the diagonal; beginning of Thibaut's proposed derivation. George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.

1.	2.	1.	11.	242.	256.
2.	8.	9.	12.	288.	289.
3.	18.	16.	13.	338.	324.
4.	32.	36.	14.	392.	400.
5.	50.	49.	15.	450.	441.
6.	72.	64.	16.	512.	529.
7.	98.	100.	17.	578.	576.
8.	128.	121.	18.	648.	625.
9.	162.	169.	19.	722.	729.
10.	200.	196.	20.	800.	784.

How far the Sútrakáras went in their experiments we are of course unable to say; the list up to twenty suffices for our purposes. Three cases occur in which the number expressing the square of the diagonal of a square differs only by one from a square-number; $8 - 9$; $50 - 49$; $288 - 289$; the last case being the most favourable, as it involves the largest numbers. The diagonal of a square, the side of which was equal to twelve, was very little shorter than seventeen ($\sqrt{289} = 17$). Would it then not be possible to reduce 17 in such a way as to render the square of the reduced number equal or almost equal to 288?

Suppose they drew a square the side of which was 17 padas long, and divided it into $17 \times 17 = 289$ small squares. If the side of the square could now be shortened by so much, that its area would contain not 289, but only 288 such small squares, then the measure of the side would be the exact measure of the diagonal of the square, the side of which is equal to 12 ($12^2 + 12^2 = 288$). When the side of the square is shortened a little, the consequence is that from two sides of the square a stripe is cut off; therefore a piece of that length had to be cut off from the side that the area of the two stripes would be equal to one of the 289 small squares. Now, as the square is composed of 17×17 squares, one of the two stripes cuts off a part of 17 small squares and the other likewise of 17, both together of 34 and since these 34 cut-off pieces are to be equal to one of the squares, the length of the piece to be cut off from the side is fixed thereby: it must be the thirty-fourth part of the side of one of the 289 small squares.

The thirty-fourth part of thirty-four small squares being cut off, one whole small square would be cut off and the area of the large square reduced exactly to 288 small squares; if it were not for one unavoidable circumstance. The two stripes which are cut off from two sides of the square, let us say the east side and the south side, intersect or overlap each other in the south-east corner and the consequence is, that from the small square in that corner not $\frac{2}{34}$ are cut off, but only $\frac{2}{34} - \frac{1}{34 \times 34}$. Thence the

Continuation of the proposed derivation: $17^2 = 289$ lies one square above $2 \times 12^2 = 288$. George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.

error in the determination of the value of the savis'esha. When the side of a square was reduced from 17 to $16\frac{33}{34}$ the area of the square of that reduced side was not 288, but $288 + \frac{1}{34 + 34}$. Or putting it in a different way: taking 12 for the side of a square, dividing each of the 12 parts into 34 parts (altogether 408) and dividing the square into the corresponding small squares, we get $408 \times 408 = 166464$. This doubled is 332928. Then taking the savis'esha-value of $16\frac{33}{34}$ for the diagonal and dividing the square of the diagonal into the small squares just described, we get $577 \times 577 = 332929$ such small squares. The difference is slight enough.

The relation of $16\frac{33}{34}$ to 12 was finally generalized into the rule: increase a measure by its third, this third by its own fourth less the thirty-fourth part of this fourth $\left(16\frac{33}{34} = 12 + \frac{12}{3} + \frac{12}{3 \times 4} - \frac{12}{3 \times 4 \times 34}\right)$

The example of the savis'esha given by commentators is indeed $16\frac{33}{34} : 12$; the case recommended itself by being the first in which the third part of a number and the fourth part of the third part were both whole numbers.

Regarding the practical use of the savis'esha, there is in Baudháyana or rather, as far as I am able to see, in all s'ulvasútras only one operation, for which it was absolutely necessary; this is, as we shall see later, the turning of a circle into a square, when the intention was to connect the rule for this operation with the rule for turning a square into a circle. A'pastamba employs (see further on) the savis'esha for the construction of right angles, but there were better methods for that purpose. The commentators indeed make the most extended use of the savis'esha, calculating by means of it the diagonals wherever diagonals come into question; this proceeding, however, is not only useless, but positively wrong, as in all such cases calculation cannot vie in accuracy with geometrical construction.

At the commencement of his sútras, Baudháyana defining the measures he is going to employ, divides the aṅguli into eight yavas, barley grains, or into thirty-four tilas (seeds of the sesame). I have no doubt that the second division which I have not elsewhere met, owns its origin to the savis'esha. The aṅguli being the measure most in use, it was convenient to have a special word for its thirty-fourth part, and to be able to say "sixteen aṅgulis, thirty-three tilas", instead of "sixteen aṅgulis, and thirty-three thirty-fourths of an aṅguli." Therefore some plant was searched for of which thirty-four seeds might be considered as equal in

Overlapping cut strips and their small corner; further historical conjectures are the editor's, not statements of the sūtra. George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.

terest to see the old ácháryas attempting this problem, which has since haunted so many unquiet minds. It is true the motives leading them to the investigation were vastly different from those of their followers in this arduous task. Theirs was not the disinterested love of research which distinguishes true science, nor the inordinate craving of undisciplined minds for the solution of riddles which reason tells us cannot be solved; theirs was simply the earnest desire to render their sacrifice in all its particulars acceptable to the gods, and to deserve the boons which the gods confer in return upon the faithful and conscientious worshipper.

It is true that they were not quite so successful in their endeavours as we might wish, and that their rules are primitive in the highest degree; but this tends at least to establish their high antiquity.

The rules are the following:

Baudháyana:

चतुरस्रं मण्डलं चिकीर्षन्नक्ष्ण्यार्धं मध्यात्प्राचीमभ्यापातयेद्यदतिशिष्यते तस्य सह तृतीयेन मण्डलं परिलिखेत्।

If you wish to turn a square into a circle, draw half of the cord stretched in the diagonal from the centre towards the práchí line (the line passing through the centre of the square and running exactly from the west towards the east); describe the circle together with the third part of that piece of the cord which will lie outside the square.

See diagram 10.

A cord is to be stretched from the centre e of the square a b c d towards the corner a; then the cord, being tied to a pole at e, is drawn towards the right hand side until it coincides in its position with the line e f; a piece of the cord, f h, will then of course lie outside the square. This piece is to be divided into three parts, and one of these three parts, f g, together with the piece e f, forms the radius of the circle, the area of which is to be equal to the area of the square a b c d.

A'pastamba gives the same rule in different words:

चतुरस्रं मण्डलं चिकीर्षन्मध्यात्कोटिं निपातयेत् पार्श्वतः परिलिख्यातिशयतृतीयेन सह मण्डलं परिलिखेत्। सा नित्या मण्डलम्। यावदीयते तावदागच्छति।

If you wish to turn a square into a circle, stretch a cord from the centre towards one of the corners, draw it round the side and describe the circle together with the third part of the piece standing over; this line gives a circle exactly as large as the square; for as much as there is cut off from the square (*viz.* the corners of the square), quite as much is added to it (*viz.* the segments of the circle, lying outside the square).

I must remark that Kapardisvámin, A'pastamba's commentator, combines the two words "sá nityá" into sánityá (= sá anityá), and explains: this line gives a circle, which is not exactly equal to the square. But I am

Square-to-circle rule attributed to Baudhāyana, followed by Āpastamba's version and discussion. George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.

afraid we should not be justified in giving to A'pastamba the benefit of this explanation. The words ' yávad dhíyate, &c.' seem to indicate that he was perfectly satisfied with the accuracy of his method and not superior, in this point, to so many circle-squarers of later times. The commentator who, with the mathematical knowledge of his time, knew that the rule was an imperfect one, preferred very naturally the interpretation which was more creditable to his author.

Kátyáyana's S'ulvaparis'ishṭa:

चतुरस्रं मण्डलं चिकीर्षन्नधाद्ँसे निपात्य पार्श्वतः परिलिख्य तच यदतिरिक्तं भवति तस्य तृतीयेन सह मण्डलं परिलिखेत्।

Let us now see what the result of the above rule would be by making the side of the square equal to 2. a c = 2; a i = 1; a e = $\sqrt{2}$ = 1·414213...; $\frac{0{\cdot}414213}{3}$ = 0·138071; radius of the circle = 1·138071.

Multiplying the square of 1·138071 by π = 3·141592..., we find as area of the circle: 4·069008....., while the area of the square = 4.

The next thing was to find a rule for turning a circle into a square. There we have at first a rule given by Baudháyana only:

मण्डलं चतुरस्रं चिकीर्षन्विष्कम्भमष्टधा भागान्कृत्वा भागमेकोनत्रिंशधा विभज्याष्टाविंशतिभागानुद्धरेद्भागस्य च षष्ठस्याष्टमभागोनम्।

If you wish to turn a circle into a square, divide the diameter into eight parts, and again one of these eight parts into twenty-nine parts; of these twenty-nine parts remove twenty-eight and moreover the sixth part (of the one left part) less the eighth part (of the sixth part).

The meaning is: $\frac{7}{8} + \frac{1}{8{\cdot}29} - \frac{1}{8{\cdot}29{\cdot}6} + \frac{1}{8{\cdot}29{\cdot}6{\cdot}8}$ of the diameter of a circle is the side of a square the area of which is equal to the area of the circle.

Considering this rule closer, we find that it is nothing but the reverse of the rule for turning a square into a circle.

It is clear, however, that the steps taken according to this latter rule could not be traced back by means of a geometrical construction; for if we have a circle given to us, nothing indicates what part of the diameter is to be taken as the "atis'ayaṭritaya" (the piece f g in diagram 10).

It was therefore necessary to express the rule for turning a square into a circle in numbers. This was done by making use of the "savis'esha", which we have considered above. Baudháyana assumed a i as equal to 12 aṅgulis (= 408 tilas), and therefore a e = 16 aṅgulis, 33 tilas. Difference = 4 aṅg. 33 til. = 169 til.; the third part of this difference = 56$\frac{1}{3}$ til. Ra-

The area mismatch, the detailed return rule and its explicit fraction expansion. George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.

dius of the circle = e f (= a i) + g f = 408 til. + 56⅓ til. = 464⅓ til. In other words: if half the side of a square is 408 til. long, the length of the radius of a circle, which is equal in area to the square, amounts to 464⅓ til.; or, if the radius of a circle is 464⅓ til., half the side of the corresponding square is 412 til. In order to avoid the fraction, both numbers were turned into thirds, and the radius made = 1393, half the side = 1224. Finally, the diameter was taken instead of the radius, and the whole side of the square instead of half the side.

To generalize this rule, it was requisite to express 1224 in terms of 1393. One eighth of 1393 = 174⅛; this multiplied by 7 = 1218⅞. Difference between 1218⅞ and 1224 = 5⅛. Dividing 174 (Baudháyana takes 174, instead of 174⅛, neglecting the fraction as either insignificant or, more likely, as inconvenient) by 29 we get 6; subtracting from 6 its sixth part we get 5 and adding to this the eighth part of the sixth part of six, we get 5⅛.

In other words: $1224 = \frac{7}{8} + \frac{1}{8 \cdot 29} - \frac{1}{8 \cdot 29 \cdot 6} + \frac{1}{8 \cdot 29 \cdot 6 \cdot 8}$ of 1393

(due allowance made for the neglected ⅛.)

Another simpler and less accurate rule for squaring the circle is common to the three Sútrakárás.

Baudháyana:

अपि वा पञ्चदश भागान्छला द्वावुदरेदेषान्नित्वा चतुरस्रकरणी।

Or else divide (the diameter) into fifteen parts and remove two; that (the remaining thirteen parts) is the gross side of the square.

A'pastamba:

मण्डलं चतुरस्रं चिकीर्षन्विष्कम्भं पञ्चदश भागान्छला द्वावुदरेन्नयोदशावशिष्यन्ते सा नित्या चतुरस्रम्।

Kátyáyana:

मण्डलं चतुरस्रं चिकीर्षन्विष्कम्भं पञ्चदश भागान्छला द्वावुदरेच्छेषः करणी।

If we assume a circle with 15 for diameter, the area of the corresponding square would, according to this rule, be 169, while the area of the circle is 176. 714.....

These are the most interesting of the paribhásha-sútras. In the following I shall extract the description of three kinds of the agnichayana, of the vakrapakshas'yenachiti, as given by A'pastamba; of the sáraratha-chakrachiti and of the s'mas'ánachiti. The two latter are described by Baudháyana only. I select these three chitis, because the first of them was, as it appears, most in use, and because some particular skill was required for the construction of the agnikshetra of the two latter chitis.

The editor's numerical link and the simpler 13/15 return rule. The prose prints 412 for the half-side, although the neighboring values use 408 and then 1224 = 3 × 408. The model follows the explicit fraction expansion on p. 27. George Thibaut, *The Śulvasūtras*, 1875. Oxford copy digitized by Google; Internet Archive ulvastras00ktgoog.